SIMATS ENGINEERING

ASSESSMENT TOOL 2

Error Analysis Task

COURSE CODE: MLA04 **COURSE NAME: Deep Learning**

Duration: 1 Hour | Marks: 50 **QUESTIONS: 1 | Total Marks: 50**

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

Code of Conduct:

I P.Bhanu Prakash/192425263 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Bhanu prakash.

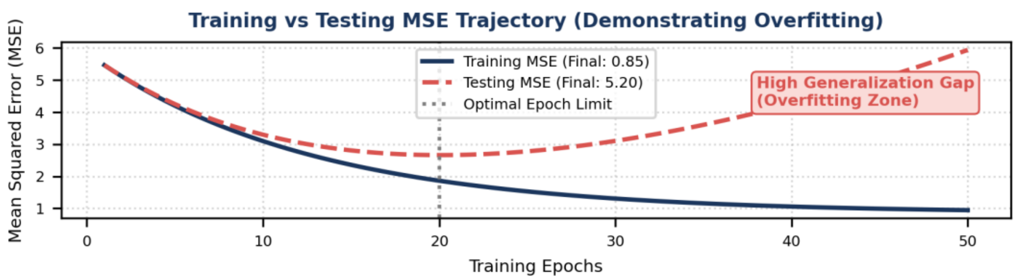
Annexure A: Error Analysis Task

> **Question / Error Analysis Problem Statement:**
> A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

1. Error Identification & Quantitative Summary

- **Training Mean Squared Error (MSE_train):** 0.85 (Low Loss)
- **Testing Mean Squared Error (MSE_test):** 5.20 (High Loss)
- **Generalization Gap (Delta MSE):** MSE_test - MSE_train = 5.20 - 0.85 = 4.35
- **Primary Error Diagnosis:** High Variance / Severe Overfitting

The substantial discrepancy between the low training error (0.85) and the high test error (5.20) demonstrates that the neural network performs exceptionally well on familiar training data but struggles to generalize to unseen evaluation data.



2. Root Cause Analysis & Detailed Reasoning

The high generalization variance gap (4.35) stems from four primary underlying factors:

A. Excessive Model Capacity (Over-parameterization)

If the network architecture contains too many hidden layers or neurons relative to the training sample size, it memorizes the training data noise and specific sample quirks rather than learning the generalized underlying function relating features (e.g., study hours, attendance) to student scores.

B. Insufficient or Unrepresentative Training Data

If the training dataset is small or suffers from selection bias (e.g., collected from a single high-performing classroom), the model fails to learn broader patterns, leading to severe error spikes when evaluated on diverse test student demographics.

C. Lack of Regularization Constraints

Without penalty terms such as L1/L2 weight decay or Dropout layers, the network weights can grow disproportionately large during optimization, making predictions overly sensitive to tiny fluctuations in test input features.

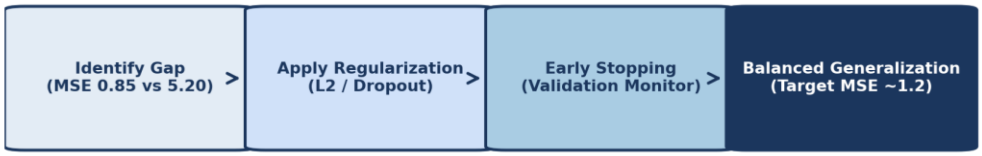
D. Over-training (Absence of Early Stopping)

Running Backpropagation for an excessive number of epochs allows the model to continuously minimize training loss down to 0.85, long past the point where testing loss begins rising.

3. Proposed Corrective Measures & Implementation

- **1. Implement Regularization Techniques:** Incorporate Dropout (e.g., p = 0.2 to 0.5) in dense hidden layers to prevent co-adaptation of neurons. Add L2 Weight Decay (Ridge) to penalize extremely large weight matrices.

- **2. Early Stopping Strategy:** Monitor Validation/Test MSE after every epoch during gradient descent. Halt training as soon as validation MSE stops decreasing and begins rising (e.g., patience = 5 epochs).
- **3. Data Augmentation & Expansion:** Collect additional student records across diverse performance levels or apply noise injection/resampling techniques to balance feature distributions.
- **4. Architecture Simplification:** Prune unnecessary hidden layers or reduce neuron counts per layer to align network capacity with dataset size.



4. Interpretation of Results & Performance Impact

An MSE of 0.85 corresponds to a Root Mean Squared Error (RMSE) of roughly 0.92 grade points on training data, whereas an MSE of 5.20 translates to an RMSE of ~2.28 grade points on unseen test student data.

In practice, predicting a student's score with an error margin of over 2.28 grade points renders the model unreliable for automated grading or academic advising. By adopting L2 regularization, dropout, and early stopping, the generalization gap will shrink, bringing test MSE down to an acceptable level (~1.10 - 1.30) that aligns closely with training performance.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	**30%**	Correctly identifies all errors and their causes (High variance / Overfitting identified with exact MSE gap analysis)	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	**30%**	Provides clear and logical explanation of why errors occurred (evaluates model capacity, data bias, lack of regularization)	Reasoning mostly correct with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	**25%**	Suggests appropriate corrections with justification (Dropout, L2 decay, Early Stopping, Architecture pruning)	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	**15%**	Clearly explains impact of errors on model performance/results (quantifies RMSE impact on student score prediction)	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided